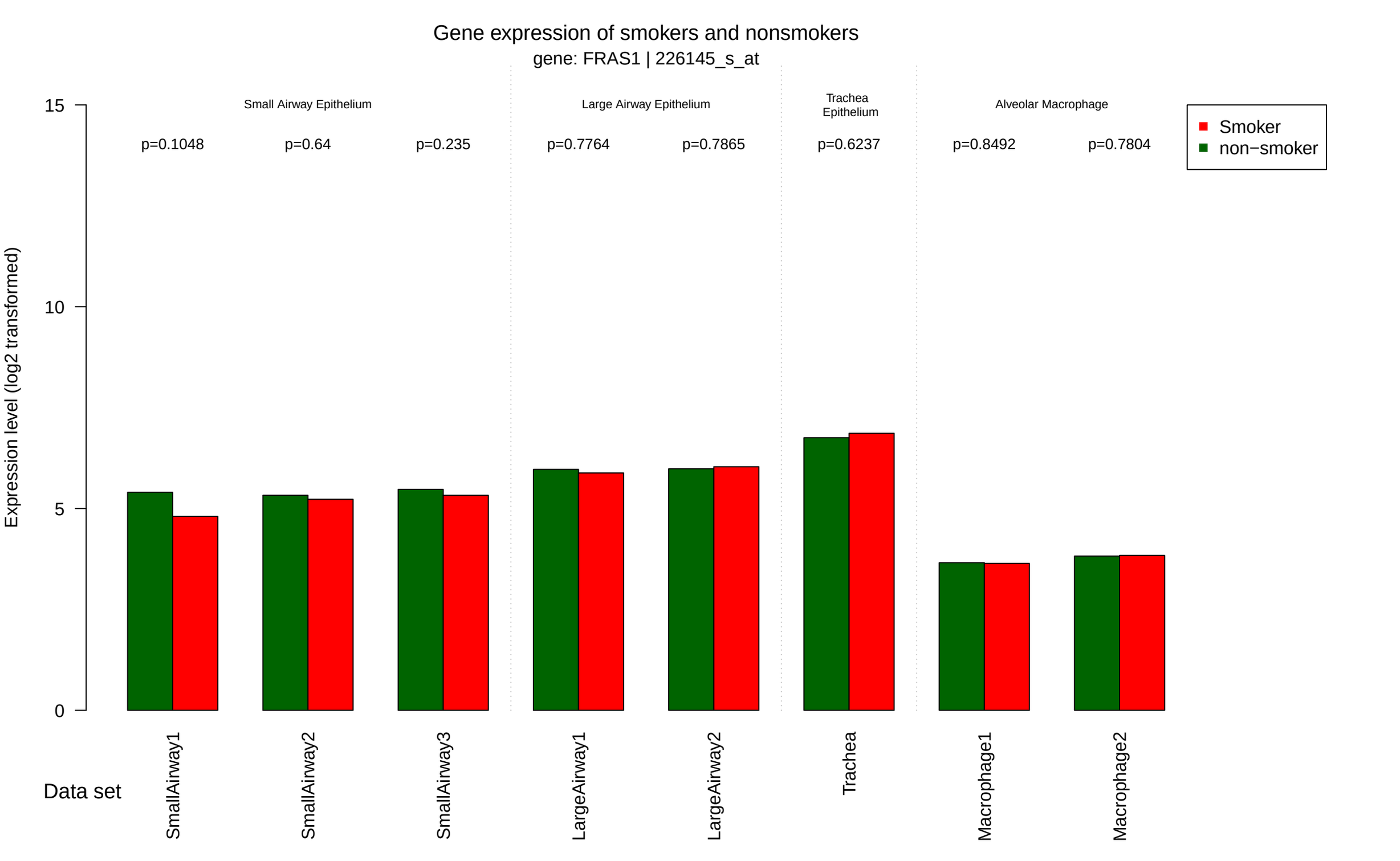
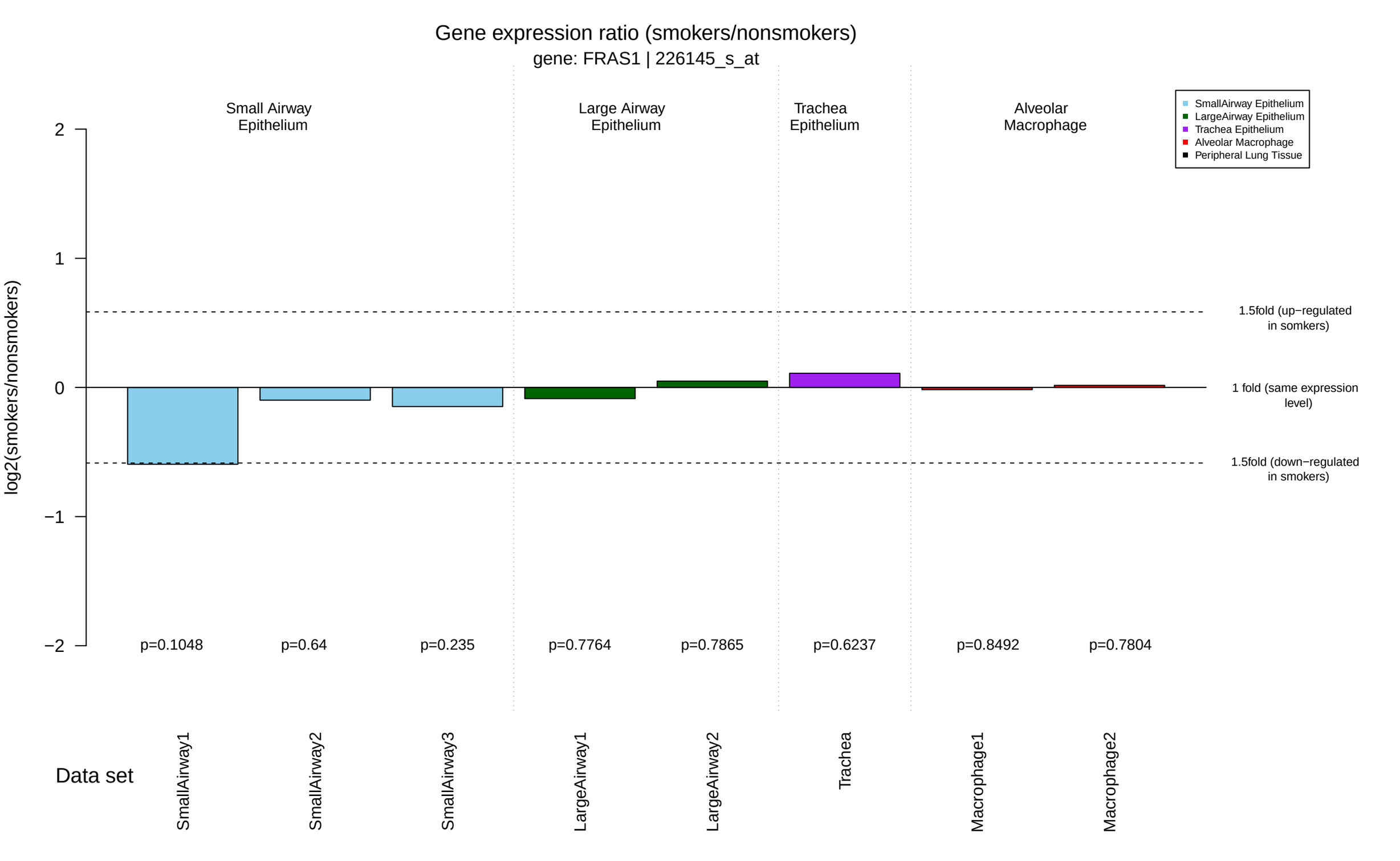
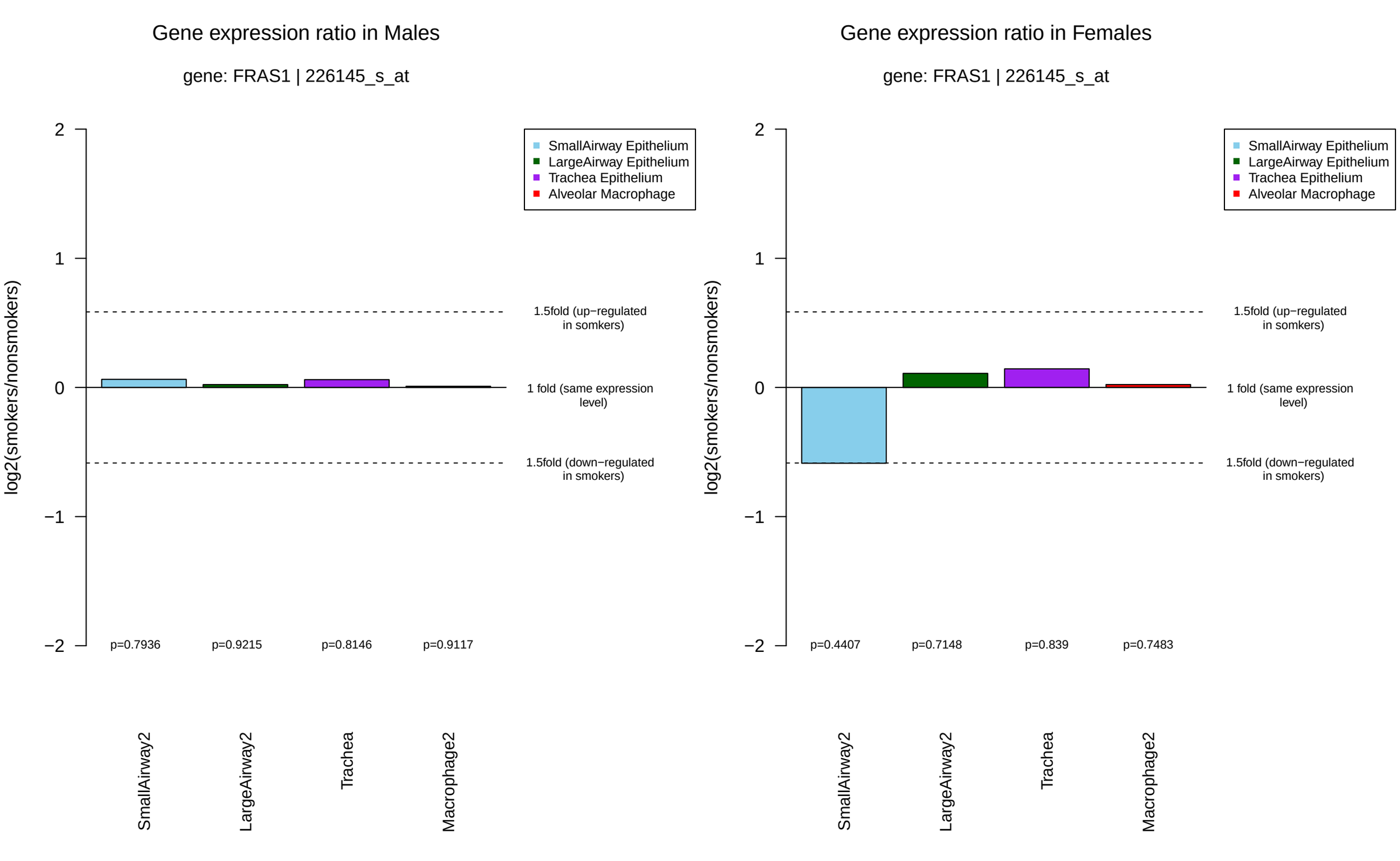
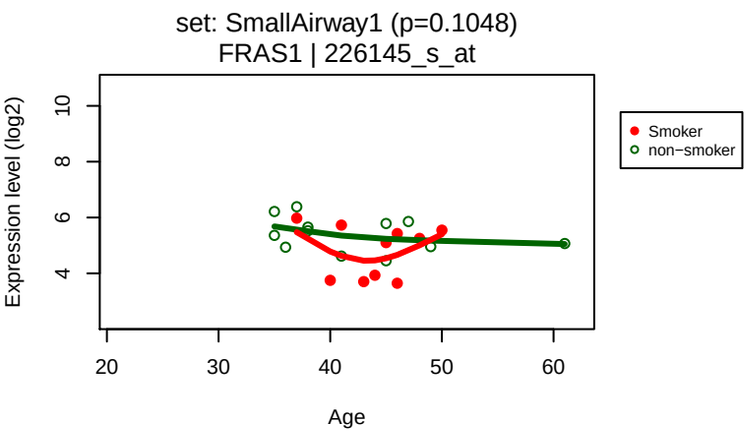
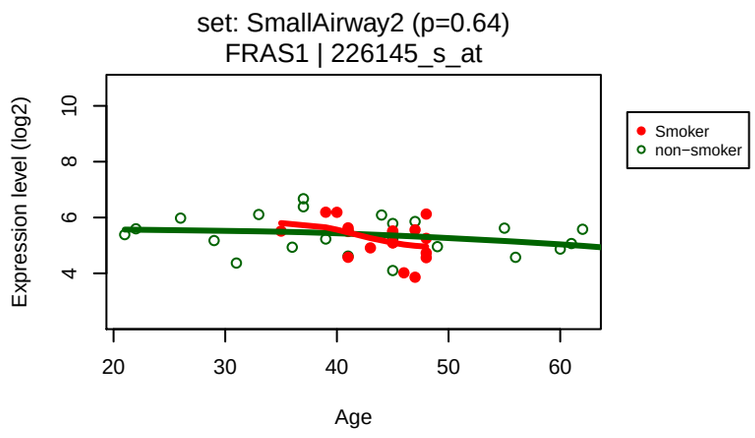
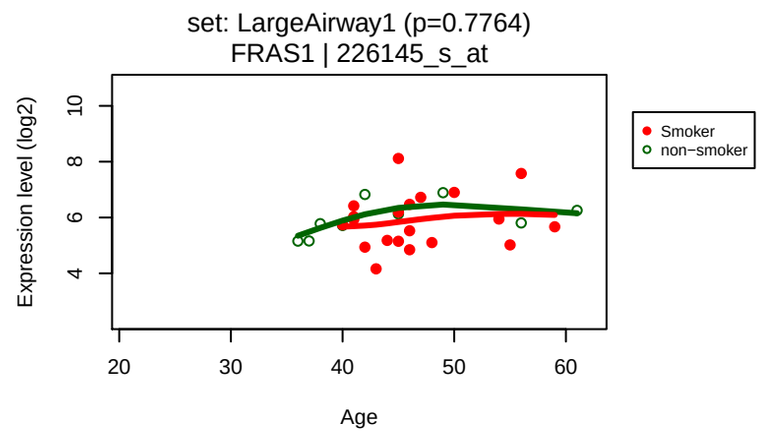

Gene expression of smokers and nonsmokers
gene: FRAS1 | 226145_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1048
p=0.64
p=0.235
p=0.7764
p=0.7865
p=0.6237
p=0.8492
p=0.7804
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

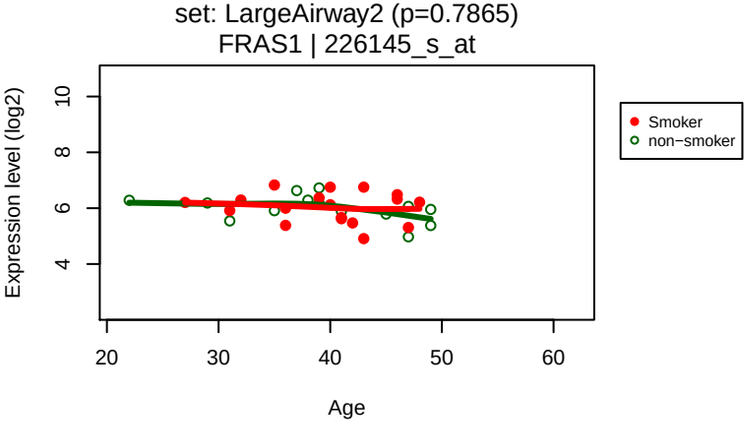

Gene expression ratio (smokers/nonsmokers)
gene: FRAS1 | 226145_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1048
p=0.64
p=0.235
p=0.7764
p=0.7865
p=0.6237
p=0.8492
p=0.7804
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

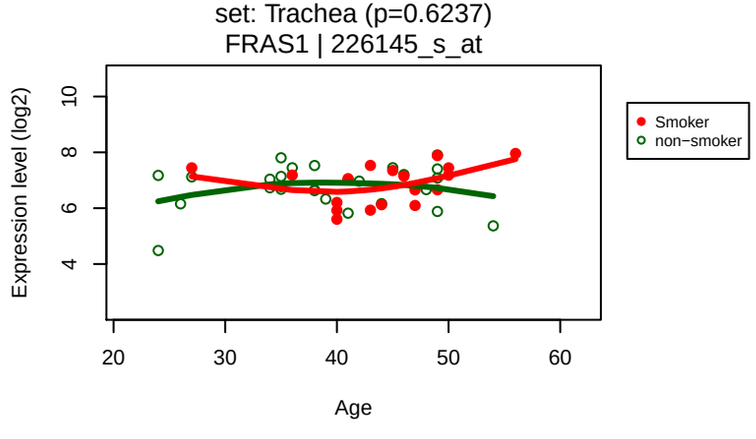

Gene expression ratio in Males
gene: FRAS1 | 226145_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7936
p=0.9215
p=0.8146
p=0.9117
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRAS1 | 226145_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4407
p=0.7148
p=0.839
p=0.7483
SmallAirway2
LargeAirway2
Trachea
Macrophage2

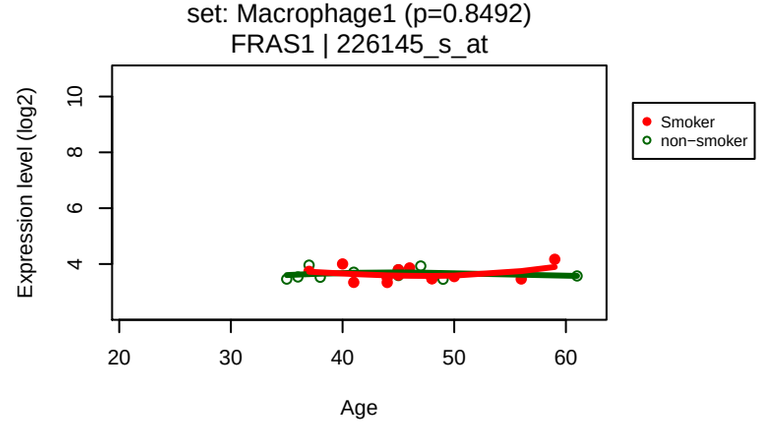

set: SmallAirway1 (p=0.1048)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker

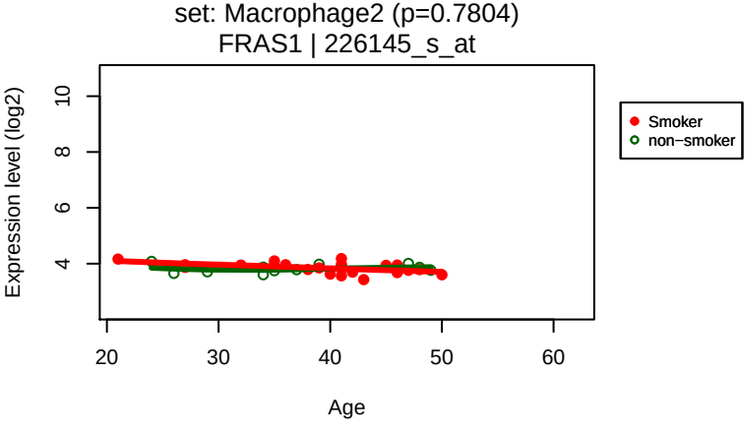

set: SmallAirway2 (p=0.64)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker

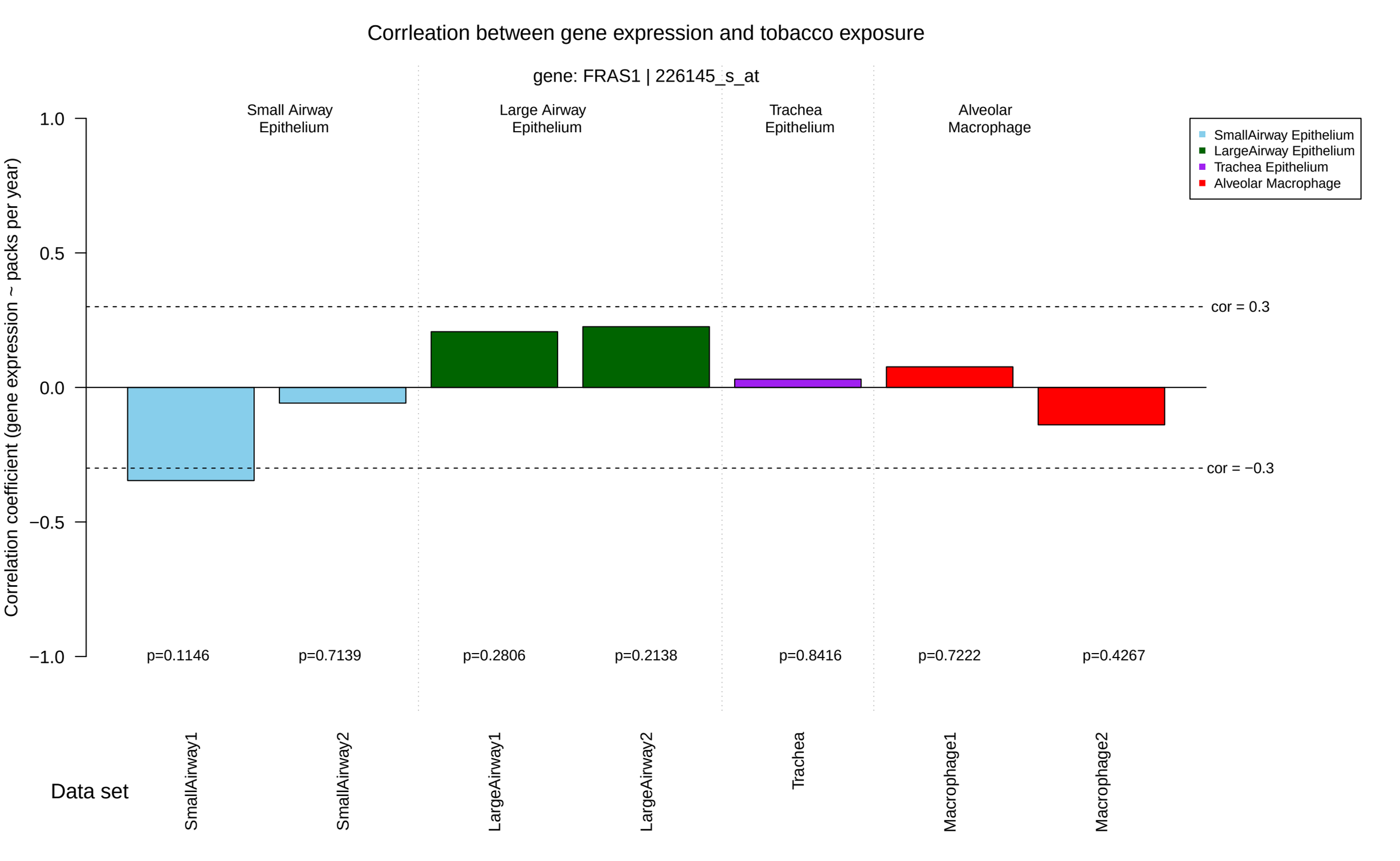

set: LargeAirway1 (p=0.7764)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7865)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6237)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8492)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7804)
FRAS1 | 226145_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRAS1 | 226145_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1146
p=0.7139
p=0.2806
p=0.2138
p=0.8416
p=0.7222
p=0.4267
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2